

AI Assisted Coding

Assignment-14.5

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Batch: 35

Lab 14 – AI Assisted HTML / CSS / JavaScript Generation

Lab Objectives

- Use AI to assist in designing and implementing fundamental data structures in Python
- Learn prompt-based structure creation, optimization, and documentation
- Improve understanding of Lists, Stacks, Queues, Linked Lists, Graphs, and Hash Tables
- Enhance readability and performance with AI-generated suggestions

Task 1 – Image Gallery Website

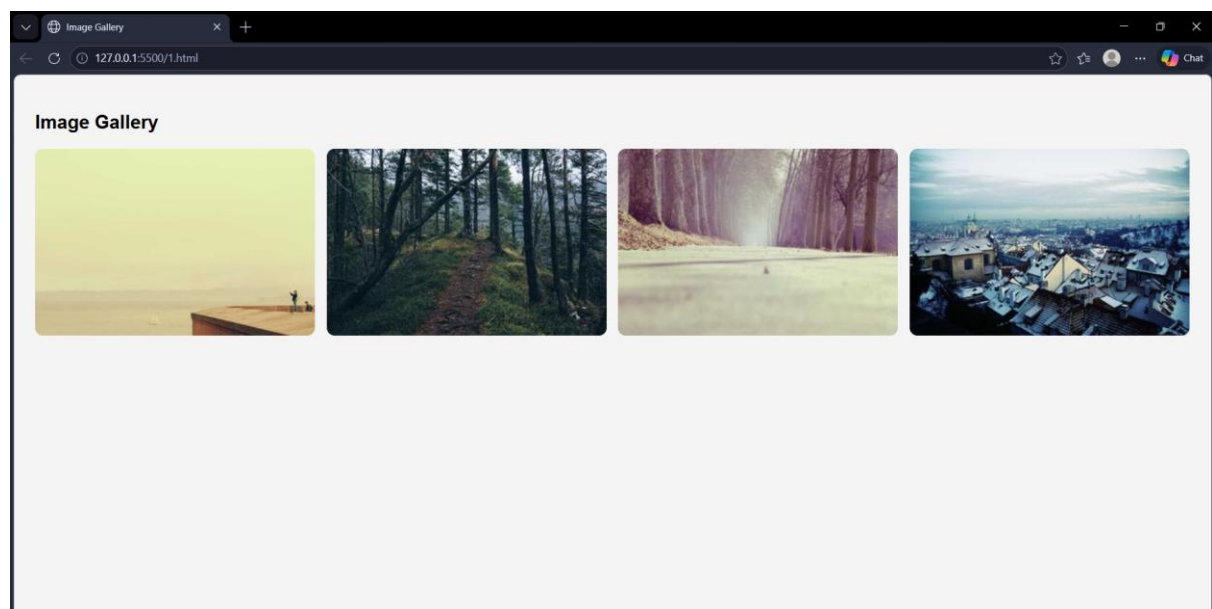
Prompt Given to AI

Generate a responsive image gallery website using HTML, CSS, and JavaScript.

Requirements:

- Display images in a grid layout using CSS Grid.
- Add hover zoom effect on images.
- Ensure the gallery is mobile responsive.
- Use clean modern UI styling.
- Provide complete code in a single HTML file.

```
1.html > html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Image Gallery</title>
5
6 <style>
7 body{
8 font-family: Arial;
9 background-color: #f4f4f4;
10 padding: 20px;
11 }
12
13 .gallery{
14 display: grid;
15 grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
16 gap: 15px;
17 }
18
19 .gallery img{
20 width: 100%;
21 border-radius: 10px;
22 transition: transform 0.3s;
23 }
24
25 .gallery img:hover{
26 transform: scale(1.1);
27 }
28 </style>
29
30 </head>
31
32 <body>
33
34 <h2>Image Gallery</h2>
35
36 <div class="gallery">
37 
38 
39 
40 
41 </div>
42
43 </body>
44 </html>
```



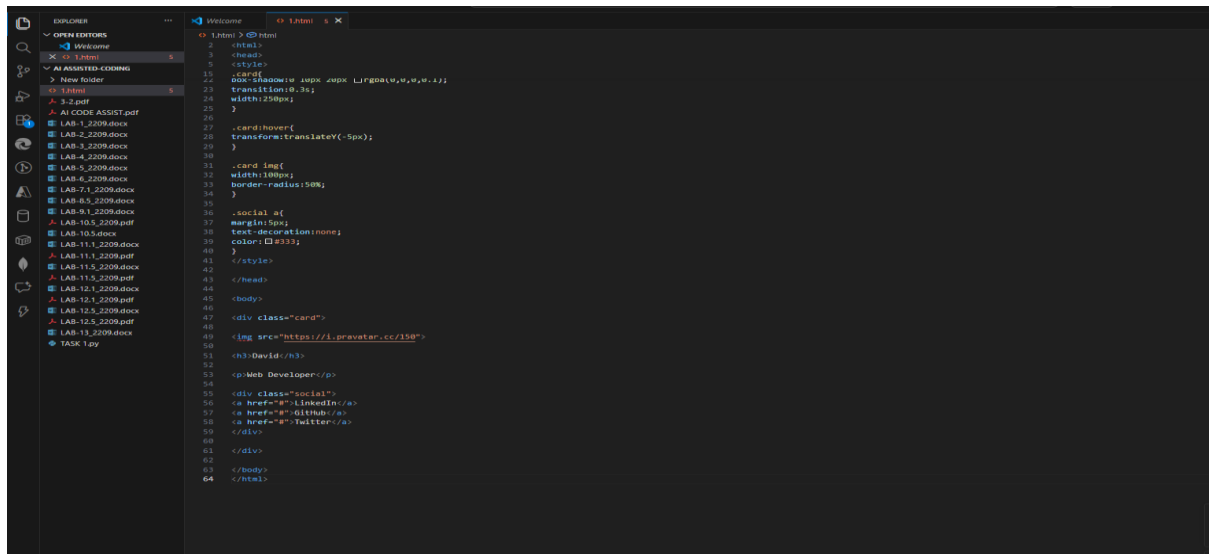
Task 2 – Profile Card Generator

Prompt Given to AI

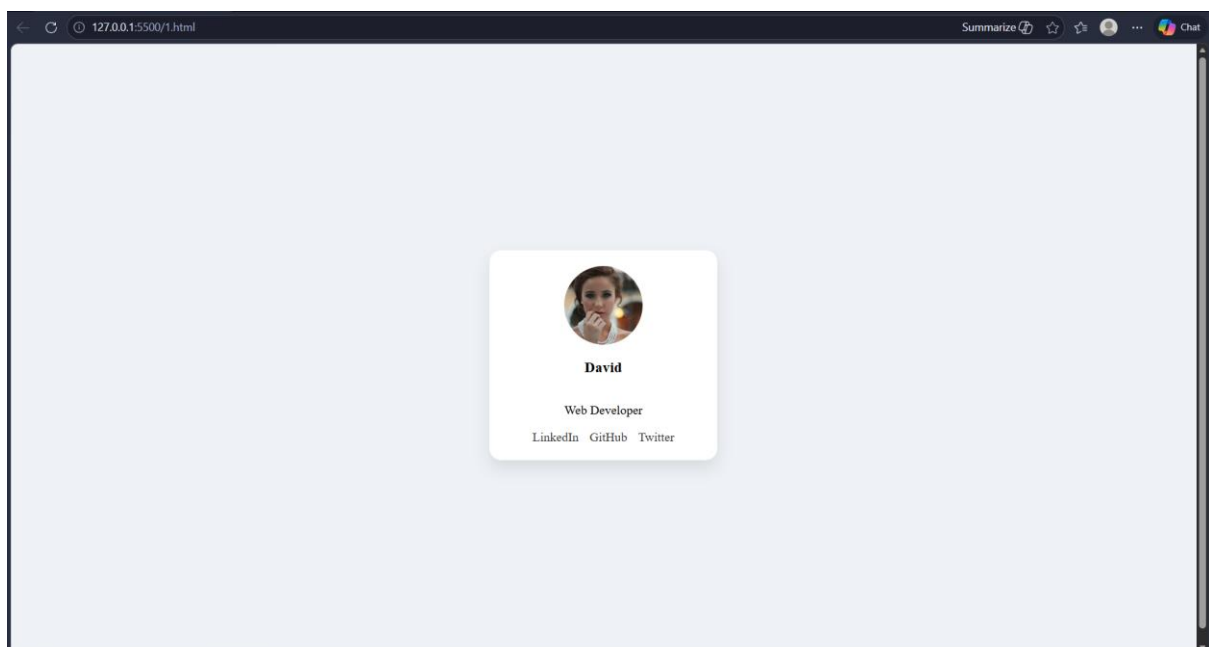
Create a modern profile card using HTML and CSS.

Requirements:

- Display profile image, name, role, and social media links.
- Use Flexbox for layout.
- Apply modern colors and fonts.
- Add hover animation and shadow effects.
- Provide the complete code.



```
1 <html>
2 <head>
3 <style>
4
5     .card {
6       box-shadow: 5px 5px 10px #f0f0f0;
7       transition: 0.3s;
8       width: 250px;
9     }
10
11     .card:hover {
12       transform: translateY(-5px);
13     }
14
15     .card img {
16       width: 100px;
17       border-radius: 50%;
18     }
19
20     .social a {
21       margin: 5px;
22       text-decoration: none;
23       color: #333;
24     }
25 </style>
26 </head>
27 <body>
28
29     <div class="card">
30       
31
32       <h3>David</h3>
33
34       <p>Web Developer</p>
35
36       <div class="social">
37         <a href="#">LinkedIn</a>
38         <a href="#">GitHub</a>
39         <a href="#">Twitter</a>
40       </div>
41     </div>
42 </body>
43 </html>
```



Task 3 – Dynamic Shopping Cart System

Prompt Given to AI

Create a dynamic shopping cart system using HTML, CSS, and JavaScript.

Requirements:

- Add and remove products.
- Update product quantities.
- Display total price dynamically.
- Use JavaScript objects to manage cart state.
- Store cart data in localStorage.
- Provide complete working code.

```
1.html > html
2  <html>
7  <body>
21 <script>
25   function addProduct(name,price){
26     cart.push({name,price})
27   }
28   localStorage.setItem("cart",JSON.stringify(cart))
29   displayCart()
30 }
31
32
33
34
35   function displayCart(){
36     let cartList=document.getElementById("cart")
37     cartList.innerHTML=""
38     let total=0
39     cart.forEach((item,index)=>{
40       let li=document.createElement("li")
41       li.innerHTML=item.name+" - "+item.price+
42       ` <button onclick="removeItem(${index})">Remove</button>`
43       cartList.appendChild(li)
44       total+=item.price
45     })
46     document.getElementById("total").innerText=total
47   }
48
49   function removeItem(index){
50     cart.splice(index,1)
51     localStorage.setItem("cart",JSON.stringify(cart))
52     displayCart()
53   }
54
55   displayCart()
56
57 </script>
58 </body>
59 </html>
```

Shopping Cart

Cart

Total: ₹0

Task 4 – Role Based Access Control

Prompt Given to AI

Create a role-based web application using HTML and JavaScript.

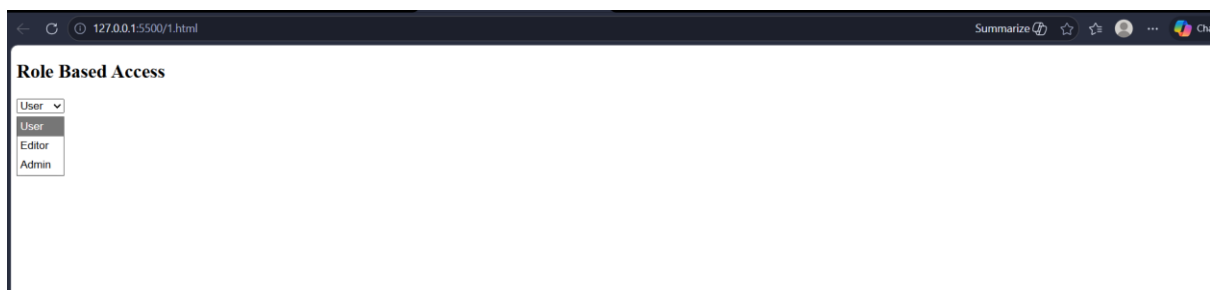
Requirements:

- Support three roles: Admin, Editor, and User.
- Display different UI components based on selected role.
- Implement conditional UI rendering.
- Provide clean and simple code.

```

1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <h2>Role Based Access</h2>
6
7 <select id="role" onchange="checkRole()">
8 <option value="user">User</option>
9 <option value="editor">Editor</option>
10 <option value="admin">Admin</option>
11 </select>
12
13 <div id="adminPanel">Admin Panel</div>
14 <div id="editorPanel">Editor Tools</div>
15
16 <script>
17
18 function checkRole(){
19
20 let role=document.getElementById("role").value
21
22 document.getElementById("adminPanel").style.display="none"
23 document.getElementById("editorPanel").style.display="none"
24
25 if(role==="admin"){
26 document.getElementById("adminPanel").style.display="block"
27 }
28
29 if(role==="editor"){
30 document.getElementById("editorPanel").style.display="block"
31 }
32
33 }
34
35 checkRole()
36
37 </script>
38
39 </body>
40 </html>

```



Task 5 – Multi Language Website

Prompt Given to AI

Create a multilingual website using HTML and JavaScript.

Requirements:

- Allow users to switch between languages dynamically.
- Store language preference using localStorage.
- Use a translation object or JSON structure.
- Provide simple example with two languages.

```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <select onchange="changeLang(this.value)">
6 <option value="en">English</option>
7 <option value="es">Spanish</option>
8 </select>
9
10 <h2 id="title"></h2>
11
12 <script>
13
14 const translations={
15   en:{title:"Welcome to the Website"},
16   es:{title:"Bienvenido al sitio web"}
17 }
18
19 function changeLang(lang){
20
21   localStorage.setItem("lang",lang)
22
23   document.getElementById("title").innerText=
24   translations[lang].title
25
26 }
27
28 let lang=localStorage.getItem("lang") || "en"
29
30 changeLang(lang)
31
32 </script>
33
34 </body>
35 </html>
```



Task 6 – Online Examination System

Prompt Given to AI

Create a simple online examination interface using HTML, CSS, and JavaScript.

Requirements:

- Display a question with options.
- Include a countdown timer.
- Automatically submit when time ends.

- Prevent multiple submissions.
- Show result after

```

1 <html>
2 <body>
3
4
5
6
7
8
9
10
11 <input type="radio" name="q1" value="3">3
12 <input type="radio" name="q1" value="4">4
13
14 <br><br>
15
16 <button onclick="submitExam()">Submit</button>
17
18 <p id="result"></p>
19
20 <script>
21
22 let time=60
23
24 let examSubmitted=false
25
26 let timer=setInterval(()=>{
27
28 time--
29
30 document.getElementById("timer").innerText="Time: "+time
31
32 if(time<=0){
33 clearInterval(timer)
34 submitExam()
35 }
36
37 },1000)
38
39 function submitExam(){
40
41 if(examSubmitted) return
42
43 examSubmitted=true
44
45 clearInterval(timer)
46
47 let answer=document.querySelector('input[name="q1"]:checked')
48
49 if(answer && answer.value=="4"){
50 document.getElementById("result").innerText="Correct"
51 }
52 else{
53 document.getElementById("result").innerText="Wrong"
54 }
55
56 }
57
58 </script>
59
60 </body>
61 </html>

```

submission.

Online Exam

Time: 38

1. What is 2 + 2?

☐ 3
☒ 4

Submit

Correct

Conclusion

AI-assisted coding tools help developers rapidly generate responsive UI layouts, optimize JavaScript logic, and implement modern web development practices. Using HTML, CSS Grid/Flexbox, and JavaScript, developers can build interactive and scalable web applications while improving usability, responsiveness, and performance.